

PVI Model

User Manual

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1. Scope

This document is a user manual of the GPT PVI DlgSILENT Model. It provides information about the main model functionalities and the necessary information for a proper use.

2. GPT PVI model description

The model is intended to be used in DlgSILENT PowerFactory version 18 (or later). It has been designed for balanced/unbalanced RMS and EMT simulations.

GPT PVI model consists on a composite model that includes the particular controls and the protections of the GP Tech SmartPV Family associated to a static generator and a shunt capacitor that represents the output AC filter reactive effect.

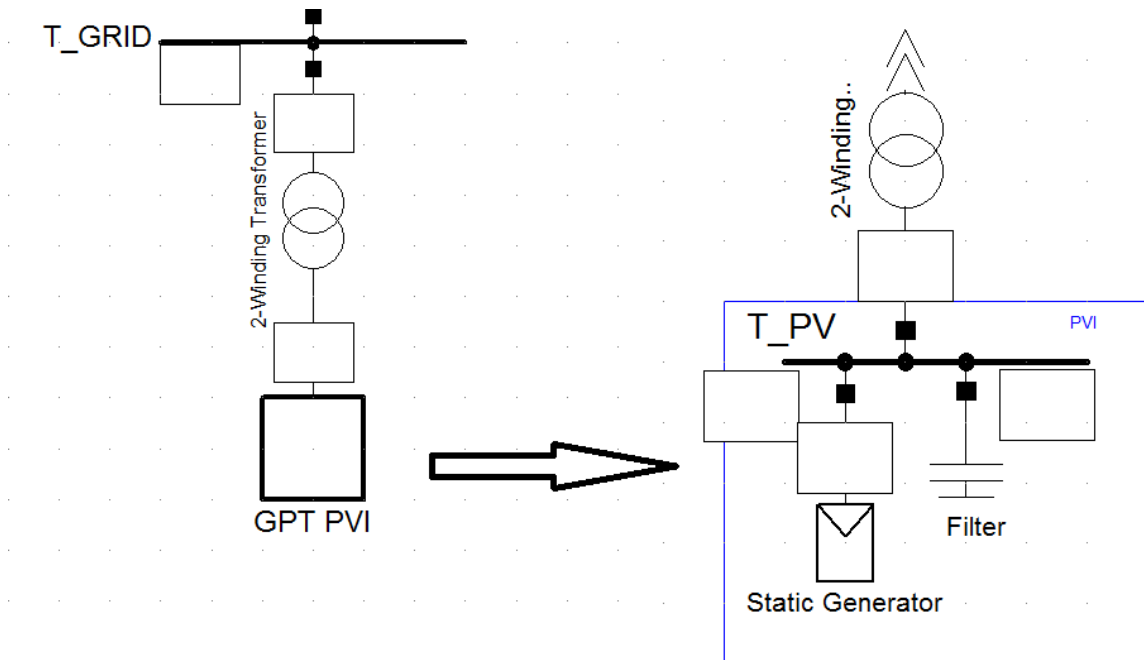


Figure 2.1. GPT PVI Model Graphical Object

The model needs input signals of irradiance, maximum active power (curtailment) and reactive power set point.

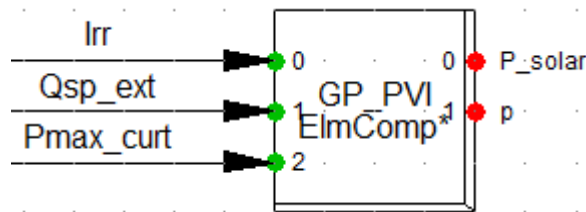


Figure 2.2. GPT PVI Model input signals

3. Functionalities

The main functionalities may be divided in two blocks: PQ Controls and Protections.

The PQ Control composite model includes:

- Local Frequency Response
- Capability Curves and P/Q priority
- Temperature derating
- Voltage Dip/Swell Control

The Protections Composite model includes:

- Over/Under Voltage
- Over/Under Frequency
- Over Current
- Asymmetrical Over Current
- Asymmetrical Over Voltage

The control algorithms are described below.

3.1 EXTERNAL SETPOINTS

The PVI model active and reactive power output shall be consistent with the values of the external control signals.

The figure below shows an example with curtailment and irradiance change events and the effect on the active power output.

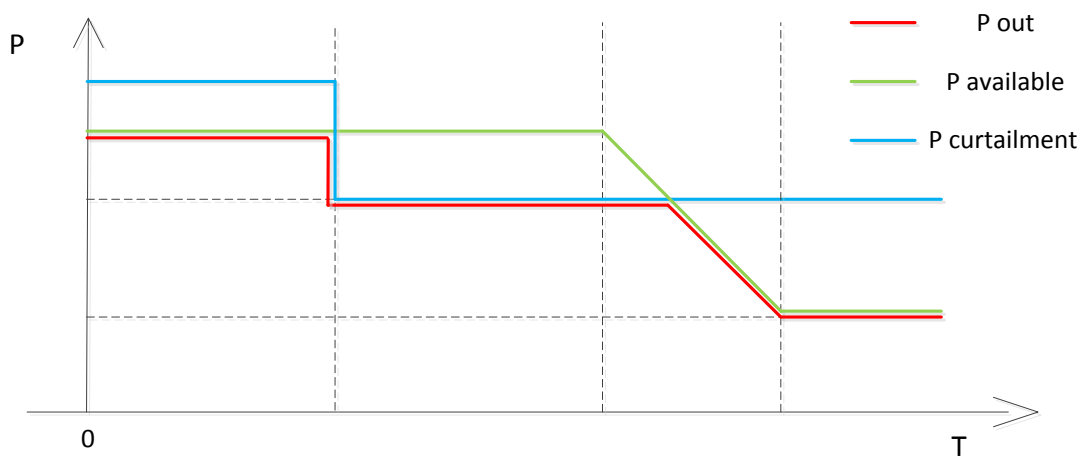


Figure 3.1. External curtailment example

3.2 LOCAL FREQUENCY RESPONSE

The frequency response control is based on an active power constraint when an over frequency event occur.

The figures below show the two types of frequency response control characteristic.

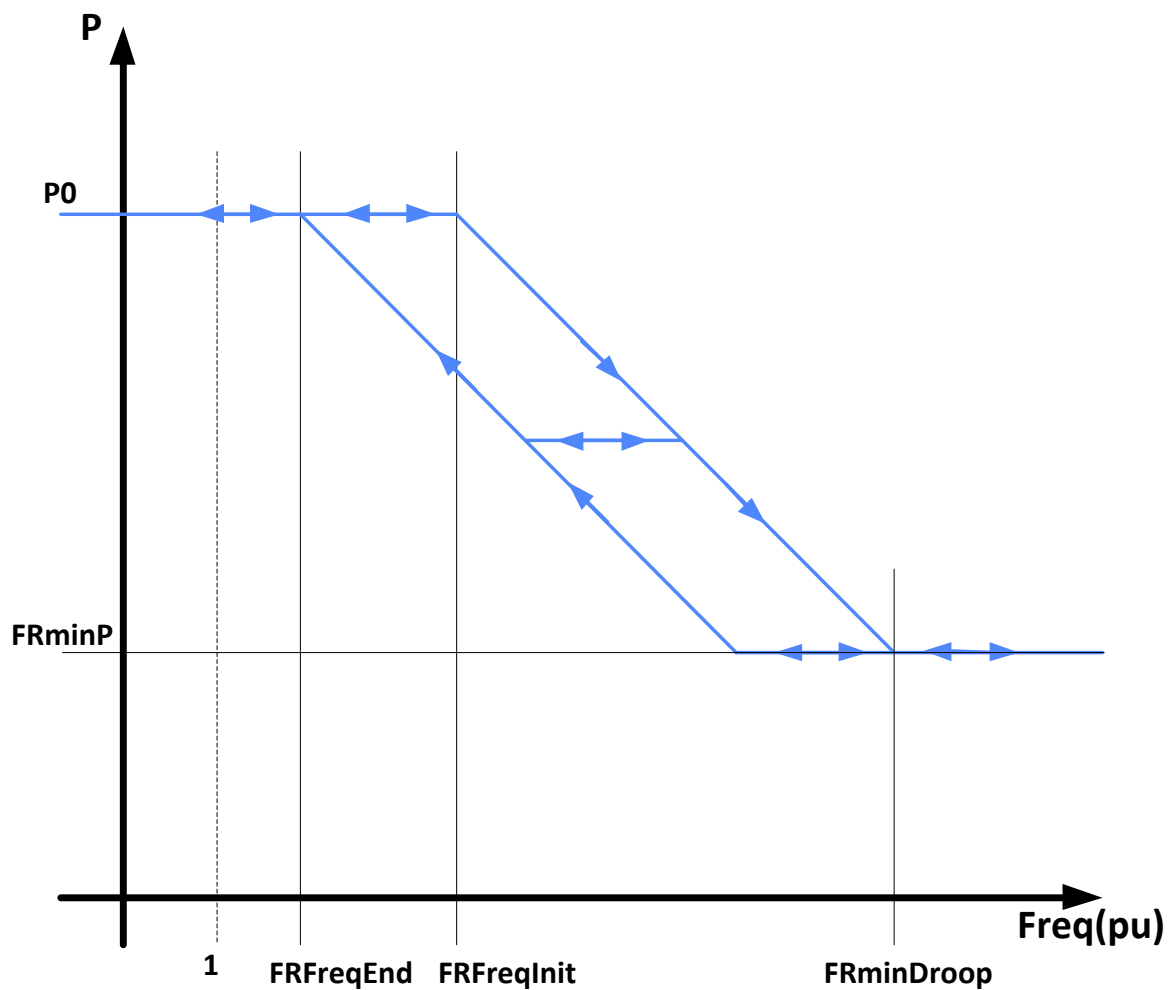


Figure 3.2. Frequency Response FRCurve 0: HYST

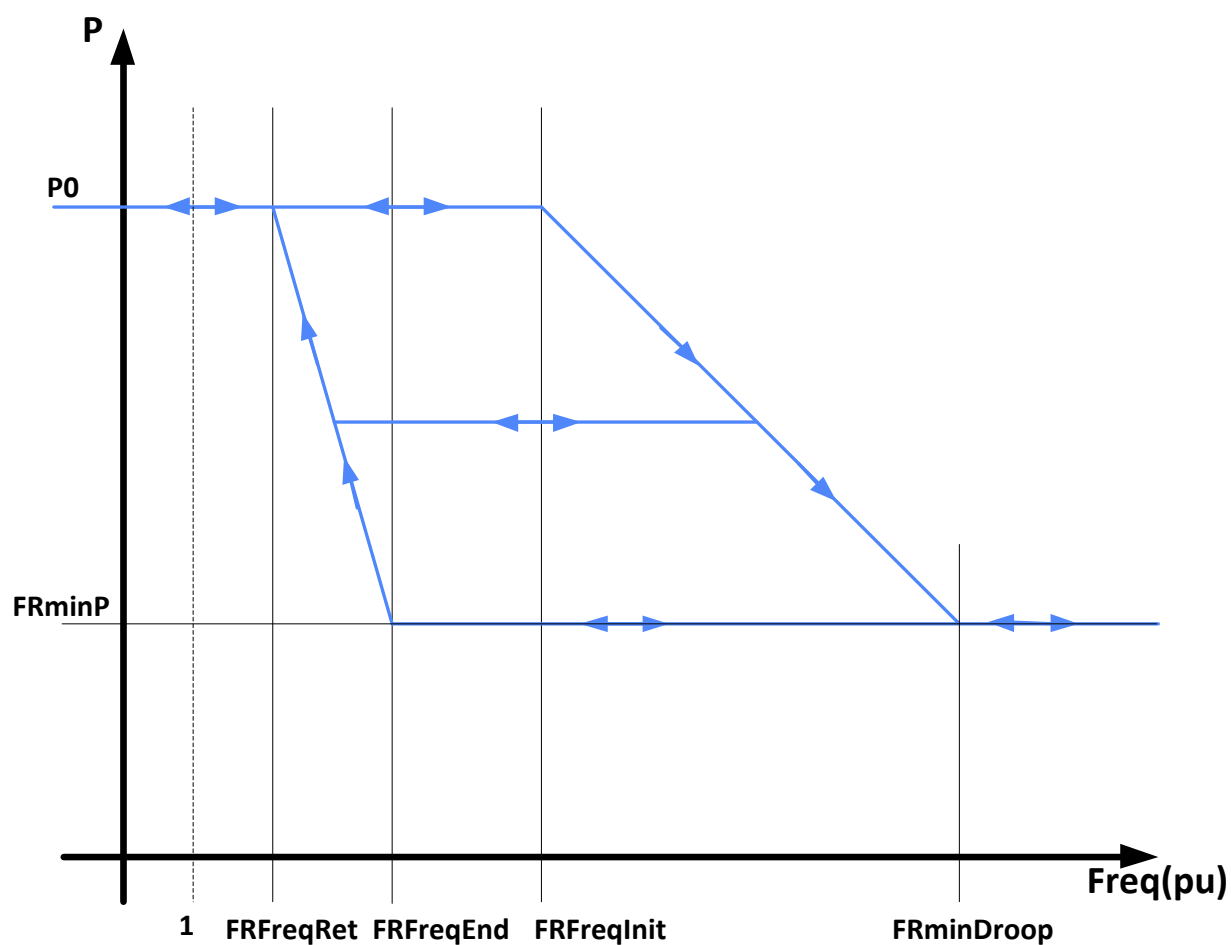


Figure 3.3. Frequency Response FRCurve 1: NO RET

3.3 CAPABILITY CURVES AND P/Q PRIORITY

The PVI capability curves are made up of I_d and I_q pairs. It shall be several curves as function of DC MPPT voltage.

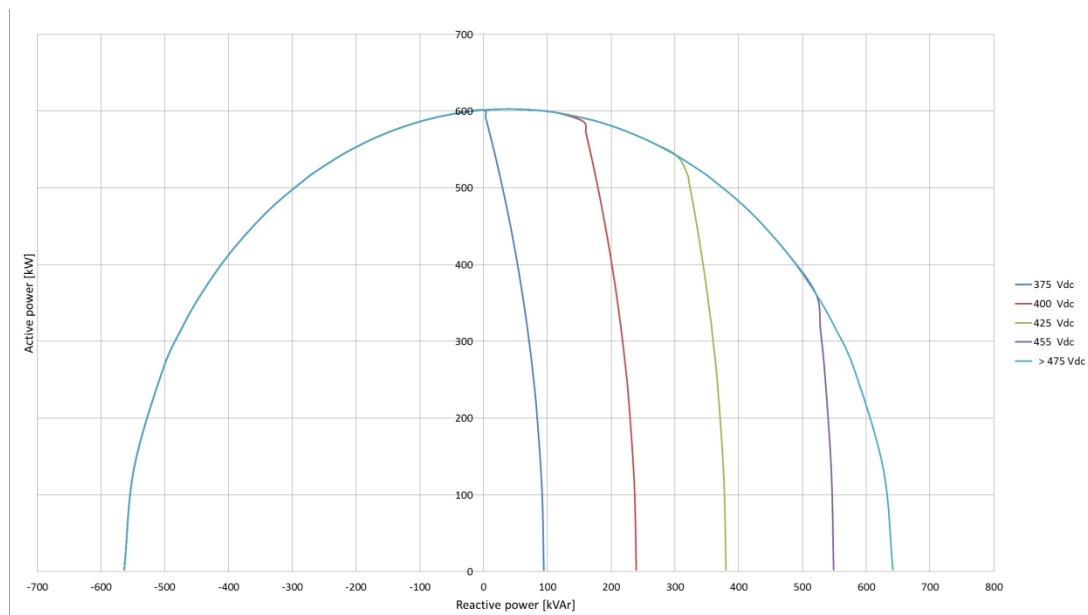


Figure 3.4. PVI capability curve example

The model also has implemented two fixed reactive power limits and one active power limit that must be defined by parameters as shown in the figure below.

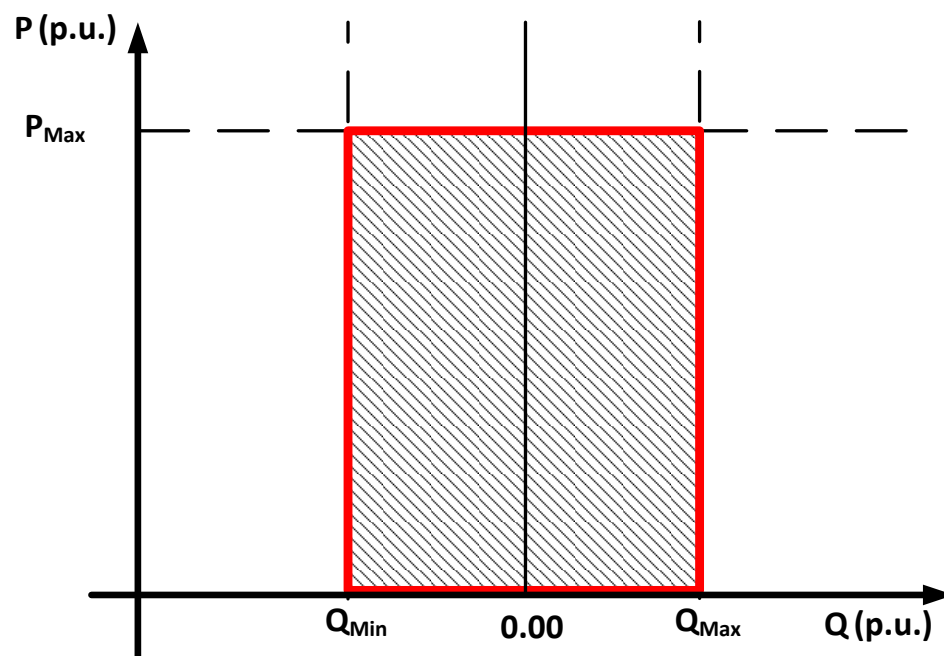


Figure 3.5. Active and reactive power fixed limits

3.4 TEMPERATURE DERATING

The maximum apparent current is limited according to a defined current versus temperature characteristic as depicted in Figure 3.6.

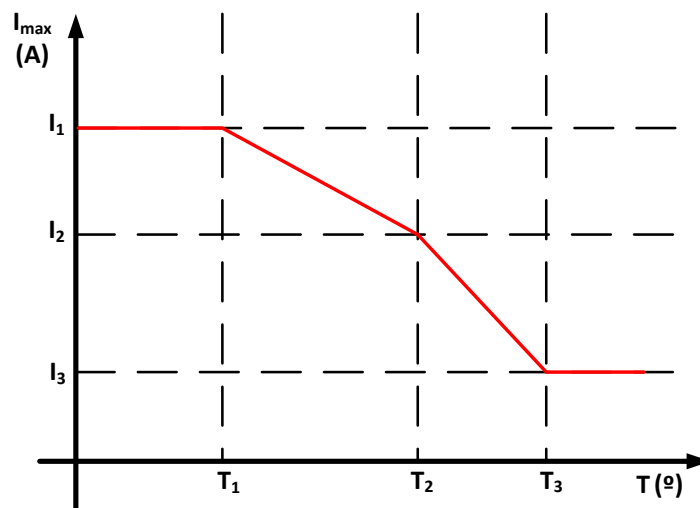


Figure 3.6. Temperature derating characteristic

3.5 VOLTAGE DIP/SWELL CONTROL

- **Dipmode = 1: INACTIVE.** The converter ceases to inject active current, only injects small reactive currents because of the filter capacitors
- **Dipmode = 2: QMAX.** The converter injects maximum reactive power allowed by defined capability curve during the zones B and C. For the low voltage ride through the converter injects the maximum capacitive current and for the high voltage ride through the converter injects the maximum inductive current.
- **Dipmode = 3: PF.** The inverter maintains the active and reactive current in a way that keeps the power factor.
- **Dipmode = 4: DROOP.** The converter injects reactive power as function of the voltage measured at his owns terminals added to the previous reactive current that was there before.

The reactive current during the zones B and C shall be according to the follow equation:

$$i_Q = i_{Qprevia} + k_{droop} (V_{mod,in} - V_{mod}) \times i_{Qmax}$$

The converter doesn't inject any active current.

- **Dipmode = 5: DROOP_MAX_P.** The converter injects reactive current as in the Dipmode = 4. The converter maintains the active power injection if possible according with the capability assumming reactive power priority.

There are multiple zones during the voltage ride through for all operating modes in which the inverter changes its behavior.

- Zone A: the converter ceases to inject current regardless of operating mode selected.
- Zone B: the converter adapts to the operating mode selected.
- Zone C: the converter keeps running as in zone B.
- Zone D: the converter ceases to inject current.
- Zone E: the converter returns to the previous operating point through a linear ramp. The value of the ramp is such that it goes from 0 to nominal current at configured time E.
- Zone F: the converter goes to normal operation according the external setpoints.

All these zones are timed through parameters as shown in the following figure.

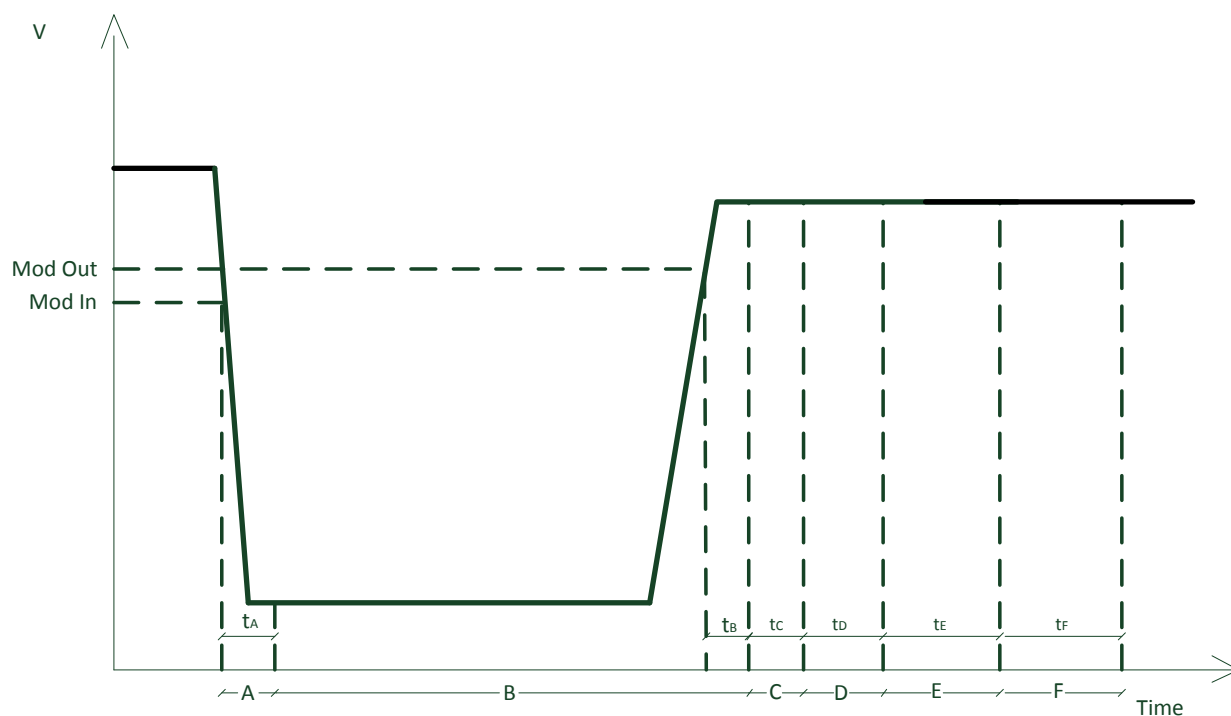


Figure 3.7. VRT zones

3.6 OVER/UNDER VOLTAGE PROTECTION

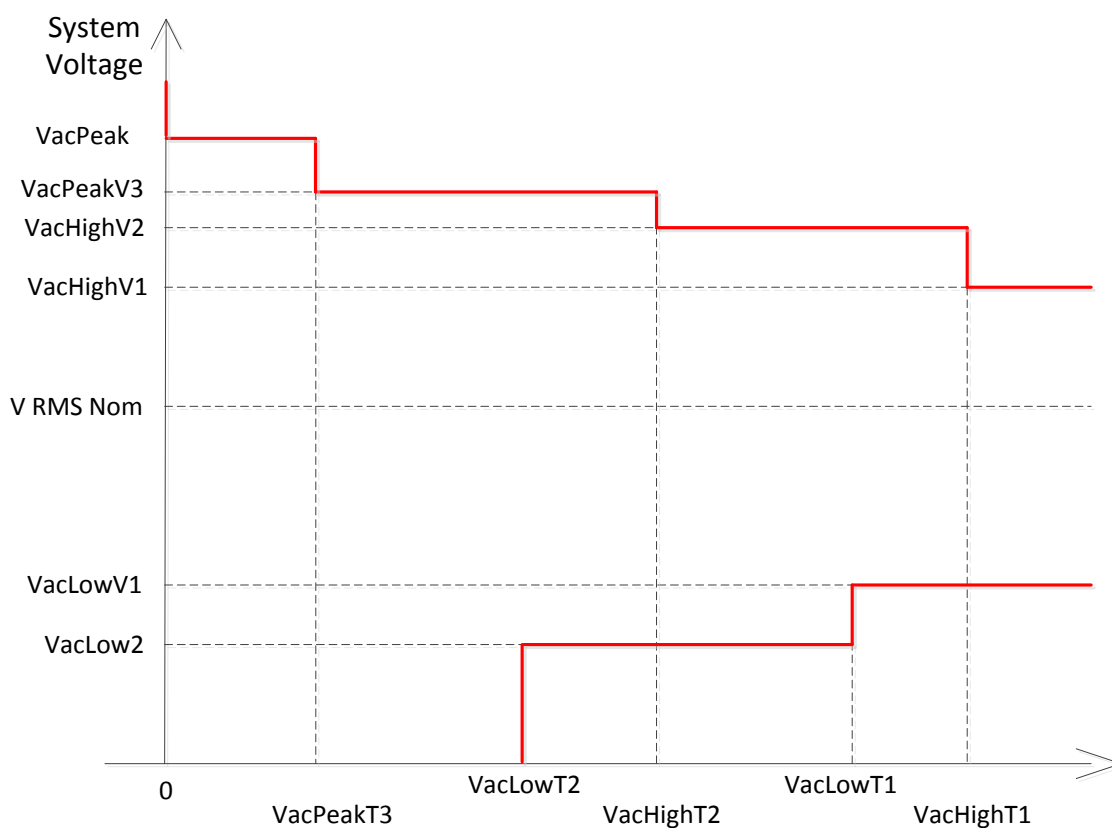


Figure 3.8. PVI over/under voltage protection characteristic

3.7 OVER/UNDER FREQUENCY PROTECTION

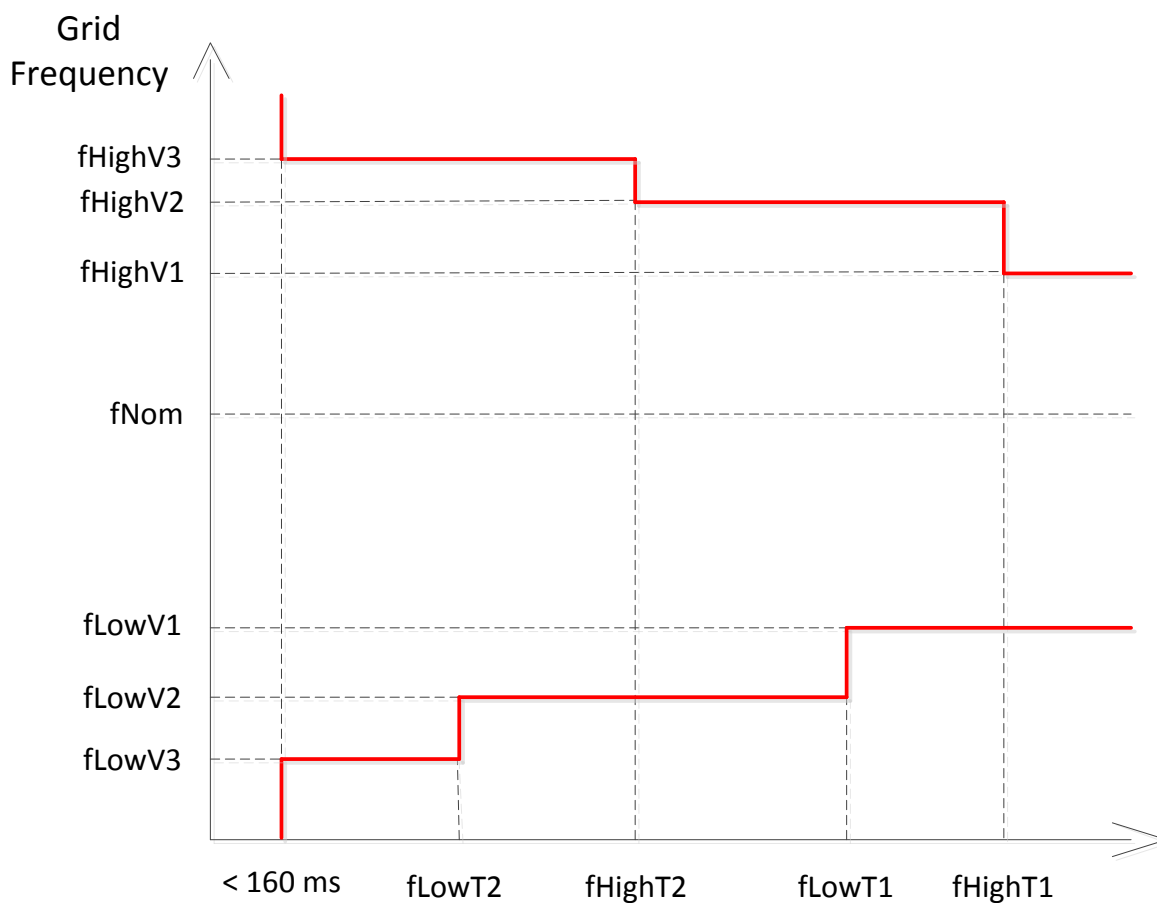


Figure 3.9. PVI over/under frequency protection characteristic

4. Use of the GPT PVI model

The deliverable of the model consists on a template containing all the necessary components to be imported in other DlgSILENT project.

It will be delivered with an example of integration. This project includes several *DlgSILENT Study Cases* that shows particular control and protection scenarios.

The model will be parameterized by GPTech personnel according to the kind of GPTech SmartPV Family unit and the customer requirements and specifications. These specifications are referred to Utility grid code for controls and protections, characteristics of the electrical facility, etc.

The configuration of the main parameters of the model will be included in the Annex I

4.1 DELIVERABLE EXAMPLE

The deliverable example consists on a DlgSILENT project where almost one GPTech PVI model has been integrated.

The GPTech PVI models are electrically connected by a step-up transformer to an equivalent grid. The project includes two types of equivalent electrical grid:

- Voltage Source and a series reactance
- Generator (Synchronous Machine) with a shunt variable load and a series reactance. It shall be activated by switching the elements only for the frequency response study cases.

The Figure 4.1 shows an example of single line diagram.

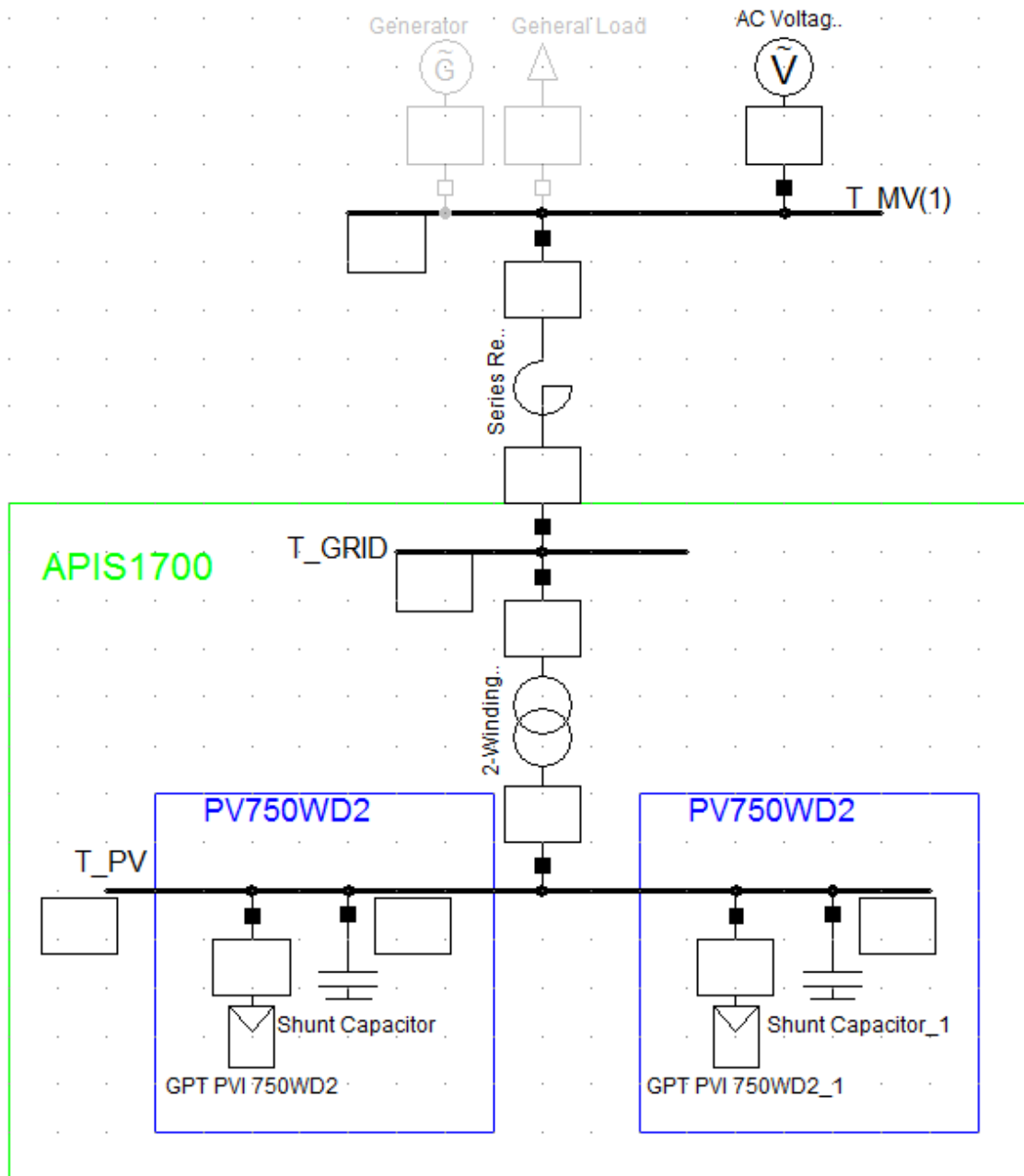


Figure 4.1. Deliverable example single line diagram

The GPTech PVI models are included in a composite model named “EXTERNAL CONTROL”.

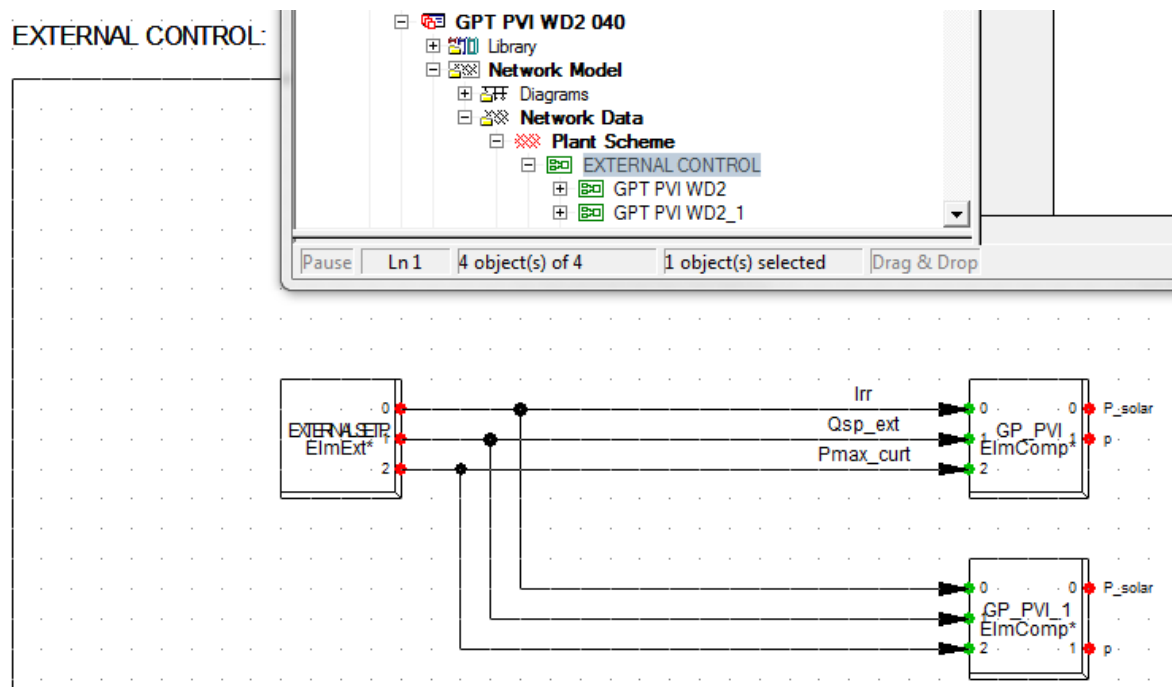


Figure 4.2. Deliverable example “EXTERNAL CONTROL” Composite Model

This composite model includes a DSL block called “EXTERNAL SETPOINTS” that allows change the input signals to the GPT PVI model as parameters.

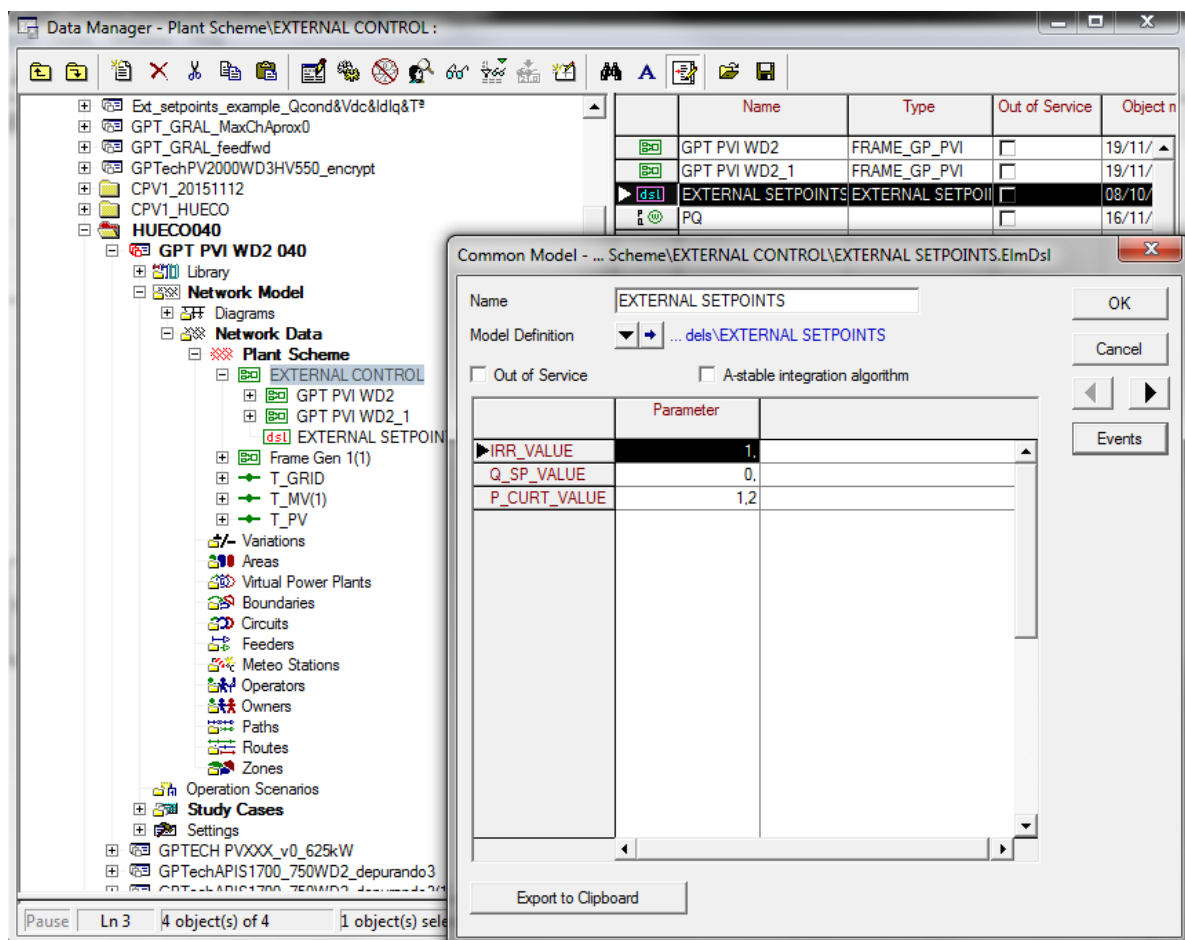


Figure 4.3. Deliverable example “EXTERNAL SETPOINTS” parameters

An example of *DigSILENT Study Case* is shown in the Figure 4.4.

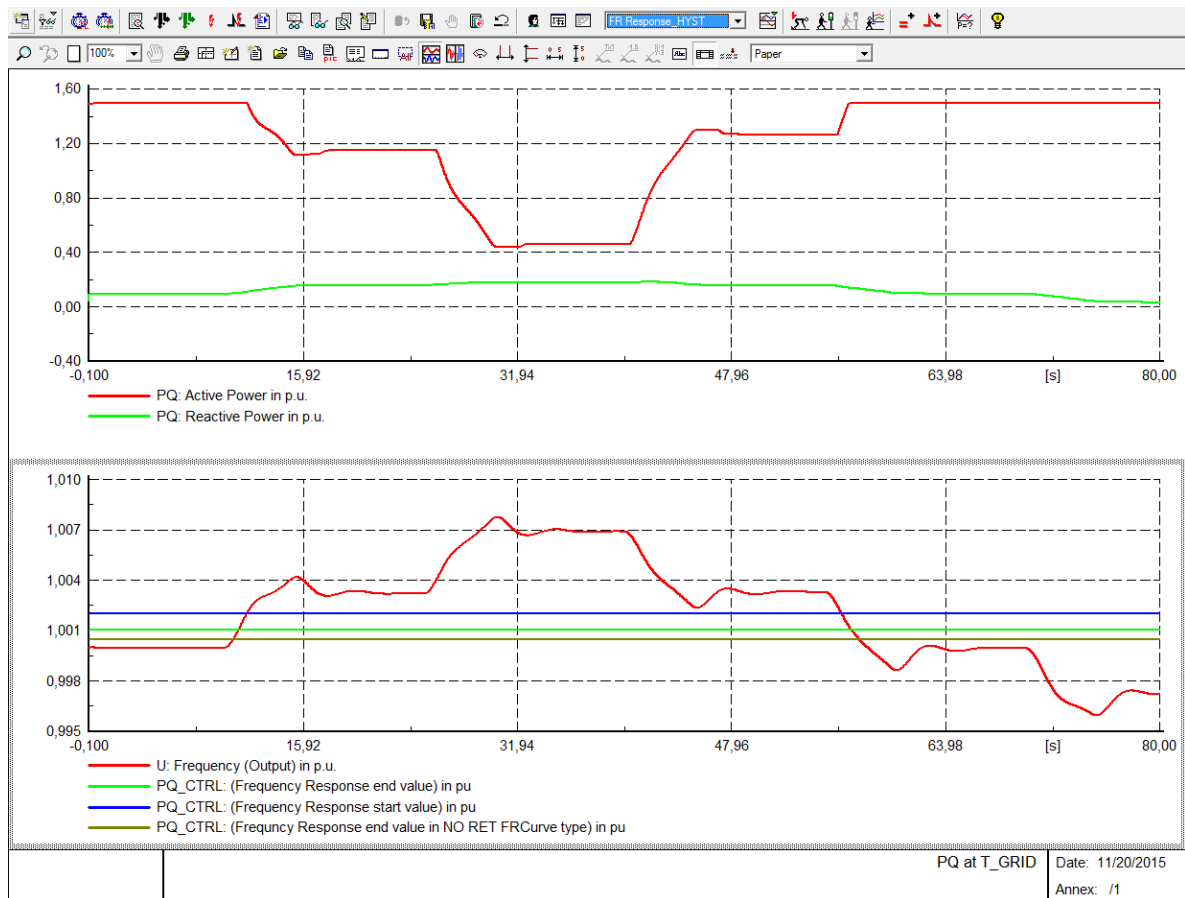


Figure 4.4. Study Case of Frequency Response example

4.2 AGGREGATED PV PLANT

The model has been designed to can be used as a whole PV plant if desired by setting up the number of parallel PV inverters as presented below.

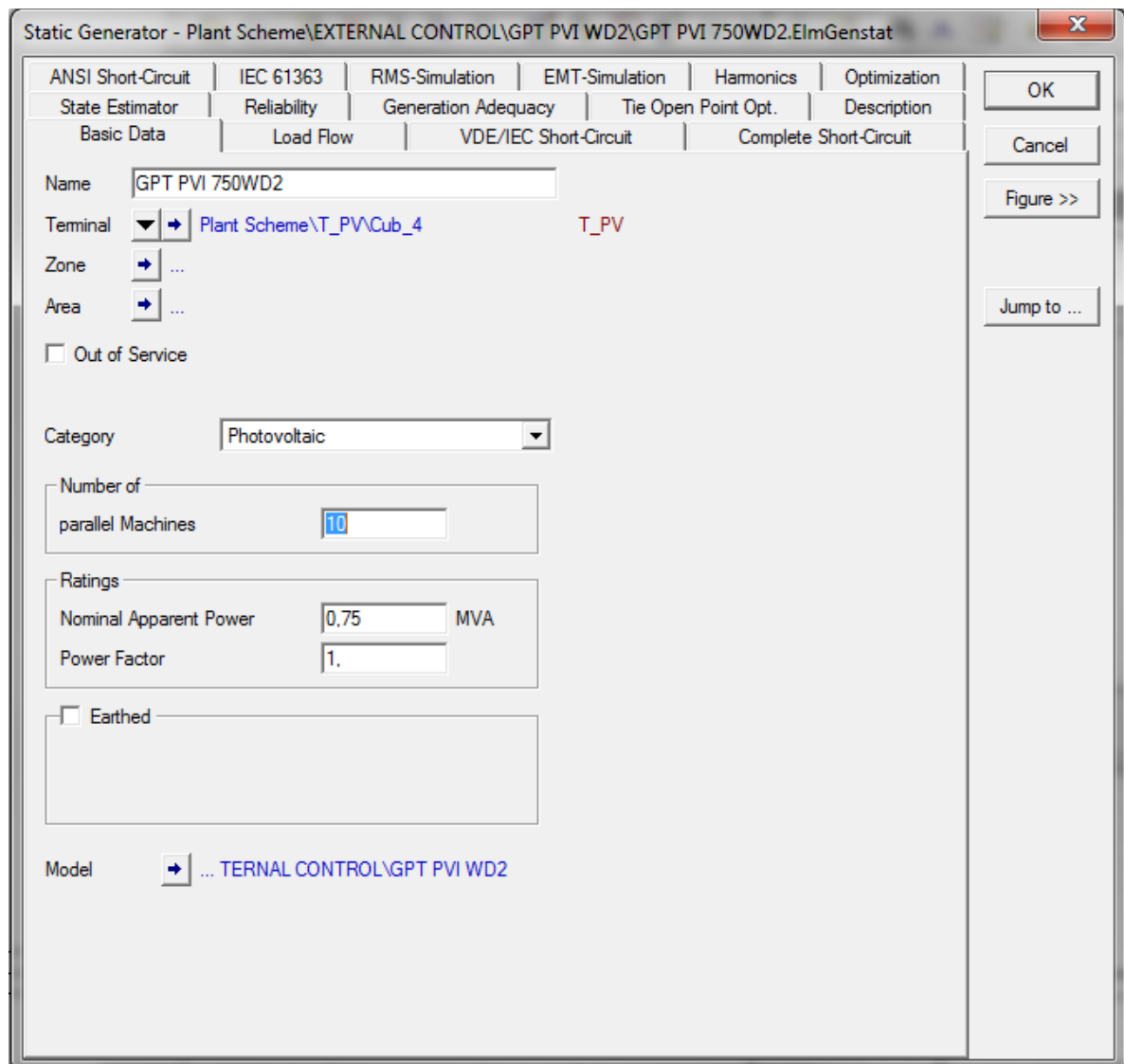


Figure 4.5. Number of parallel machines for aggregated model

The number of parallel transformers must be modified, as in Figure 4.6.

2-Winding Transformer - Plant Scheme\2-Winding Transformer.ElmTr2

RMS-Simulation	EMT-Simulation	Hamonics	Optimization	State Estimator
Reliability	Generation Adequacy	Tie Open Point Opt.		Description
Basic Data	Load Flow	VDE/IEC Short-Circuit	Complete Short-Circuit	ANSI Short-Circuit IEC 61363

Name: 2-Winding Transformer

Type: ... ent Type Library\2-Winding Transformer Type(1)

HV-Side: Plant Scheme\T_GRID\Cub_4 T_GRID

LV-Side: Plant Scheme\T_PV\Cub_1 T_PV

Zone: HV-Side

Area: HV-Side

☐ Out of Service ☐ External Star Point

Number of parallel Transformers: 10

Flip Connections

Thermal Rating: ...

Rating Factor: 1. Rated Power: 1,7 MVA

Internal Grounding Impedance, HV Side

Star Point: Connected

☐ Petersen Coil

Resistance, Re: 0. Ohm

Reactance, Xe: 0. Ohm

OK

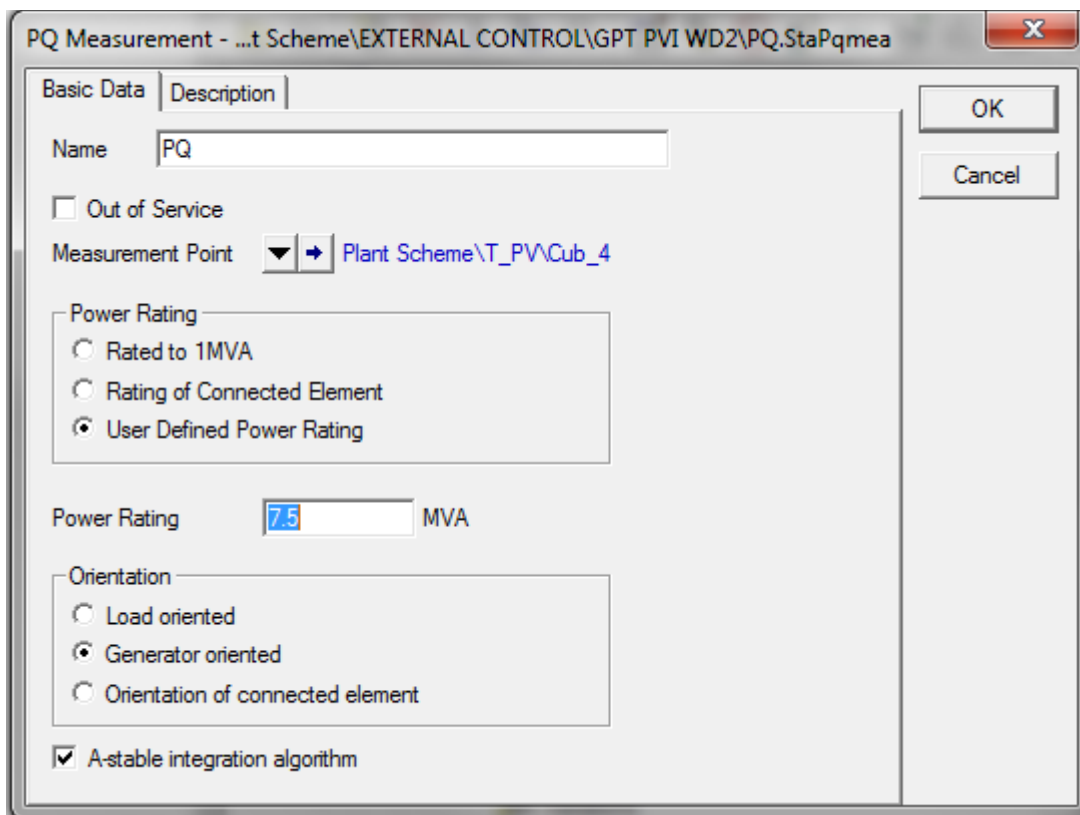
Cancel

Figure >>

Jump to ...

Figure 4.6. Number of parallel transformers for aggregated model

The power measurement device must be adapted to the new power flow as presented in the figure below.



PQ Measurement - ...t Scheme\EXTERNAL CONTROL\GPT PVI WD2\PQ.StaPqmea

Basic Data | Description

Name: PQ

☐ Out of Service

Measurement Point: ▼ ➡ Plant Scheme\T_PV\Cub_4

Power Rating

☐ Rated to 1MVA

☐ Rating of Connected Element

☒ User Defined Power Rating

Power Rating: 7.5 MVA

Orientation

☐ Load oriented

☒ Generator oriented

☐ Orientation of connected element

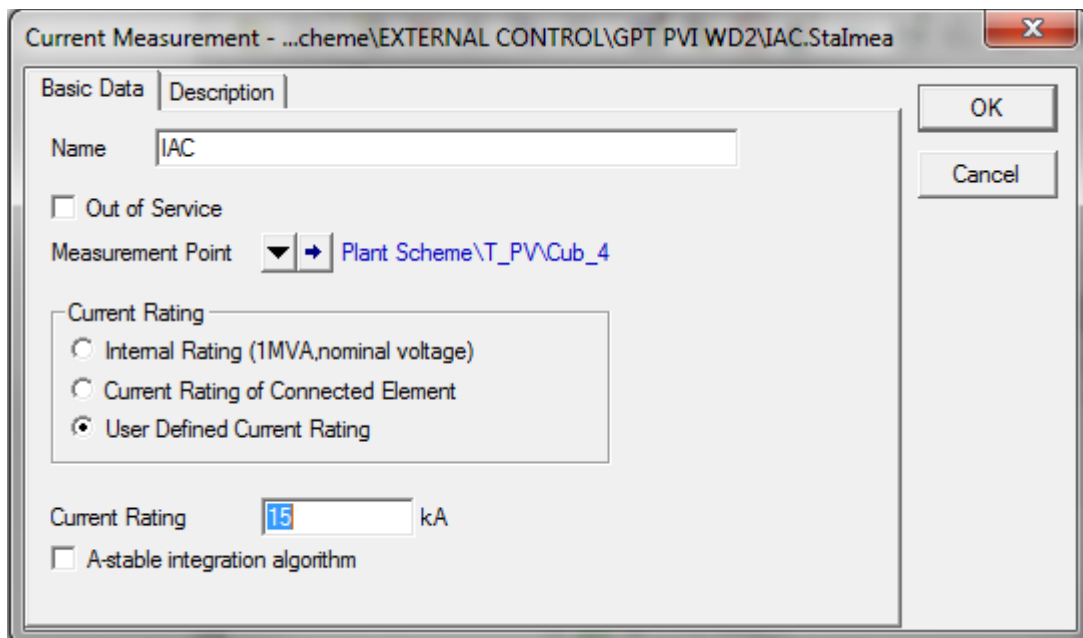
☒ A-stable integration algorithm

OK

Cancel

Figure 4.7. Power measurement setting for aggregated model

The current measurement device must be adapted to the new power flow as presented in the figure below.



Current Measurement - ...cheme\EXTERNAL CONTROL\GPT PVI WD2\IAC.StaImea

Basic Data | Description

Name: IAC

☐ Out of Service

Measurement Point: ▼ → Plant Scheme\T_PV\Cub_4

Current Rating

- ☐ Internal Rating (1MVA,nominal voltage)
- ☐ Current Rating of Connected Element
- ☒ User Defined Current Rating

Current Rating: 15 kA

☐ A-stable integration algorithm

OK Cancel

Figure 4.8 Current measurement setting for aggregated model

5. Contact

If you require more information, please contact GPTech through the website (www.greenpower.es) or through the following addresses:

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6. Document Revisions

REVISION TABLE					
Revision	Date	Elaborated	Reviewed	Approved	Modification: page, number, paragraph, description
0.1.0	20-11-2015	JR Gordillo			First release
0.1.1	30-11-2015	JR Gordillo			Scope and description modification
0.1.2	23-05-2017	JR Gordillo			Functionalities description added
0.1.3	22-09-2017	JR Gordillo			Freq Response Figures revision
0.1.4	25-01-2019	JR Gordillo			EMT simulations allowed

ANNEX I. Model parameterization

General		
Inverter Station	--	
PVI	PV3xMWD3-HV660	
Step-up transformer	Aparent power	4020 kVA
	Type	1HV+1LV Winding Power Transformer
	Voltages	20/0.66
	Xcc	HV-LV 7 %
		LV-LV --
	No load losses	--
PQ Controls		
Frequency Response	FREnable (0 : DISABLE / 1 : ENABLE) [0/1]	0
	FRCurve (0 : HYST / 1 : NO RET) [0/1]	--
	FRminP (Minimum Active Power due to frequency Response) [pu]	--
	FRFreqInit (Frequency Response start value) [pu]	--
	FRFreqEnd (Frequency Response end value) [pu]	--
	FRKDrop (DeltaP(Freq) Characteristic Droop) [pu(W)/pu(Hz)]	--
	FRFreqRet (Frequency Response end value in NO RET FRCurve type) [pu]	--
	FRRampRet (Ramp rate during the active power recovery) [pu/sec]	--
	FRtWaitRet (Delay time to start the active power recovery) [sec]	--

	FRminDroop (Frequency value that defines the maximum power decrease) [pu]		--
Capability	PQPRIORITY (0 : priority P / 1 : priority Q) [0/1]		0
	Pmax (Maximum Active Power) [pu]		1
	Qmax (Maximum Capacitive Power) [pu]		1
	Qmin (Maximum InductivePower) [pu]		-1
	Temperature (Ambient temperature) [°C]		35
	VdcMPPT (MPPT DC Voltage) [Vdc]		1070
	Array TempCurrent (Active Power Derating) [°C Amp]	-10	1172
		35	1172
		50	1050
		55	910
		60	760
	Array VdcMPPT (Capability as function of Vdc) [Vdc Cap Curve]	967	0
		1010	1
		1030	2
1050		3	
1070		4	
Voltage Dip/Swell Control	Dipmode (0 : DISABLED / 1 : INACTIVE / 2 : QMAX / 3 : PF / 4 : DROOP / 5 : DROOP_MAX_P) [0/.../5]		5
	UpDipmode (0 : DISABLED / 1 : ENABLED) [0/1]		1
	IsComp (Maximum aparent current) [pu]		1
	modIn (Dip initial voltage) [pu]		0,89
	modOut (Dip final voltage) [pu]		0,93
	UpperModIn (Swell initial voltage) [pu]		1,11
	UpperModOut (Swell final voltage) [pu]		1,07

TimeA (Time in Zone A) [sec]		0,1
TimeB (Time in Zone B) [sec]		0,5
TimeC (Time in Zone C) [sec]		0
TimeD (Time in Zone D) [sec]		0
TimeE (Time in Zone E) [sec]		1
TimeF (Time in Zone F) [sec]		0
Array dip_profile (Time delayed protection during a voltage dip) [pu sec]	0	0,26
	0,3	1,01
	0,8	3,01
	1	10,01
Array swell_profile (Time delayed protection during a voltage swell) [pu time sec]	1,1	1000
	1,11	2
	1,2	2
	1,21	1
	1,3	1
	1,31	0,02
	1,31	0,02
	1,31	0,02
Array u_deltaIq (deltaIq characteristic as function of Vac) [pu Amp]	0	-1
	0,4	-1
	0,9	0
	1,1	0
	1,4	0,6
	1,4	0,6
	1,4	0,6

Protections		
Over voltage Protection	ONOFF_OV (0:off/1:on) [0/1]	0
	Level_OV (Overvoltage detection) [pu]	--
	Array levelOV_time () [pu sec]	--
		--
		--
		--
		--
		--
Under voltage Protection	ONOFF_UV (0:off/1:on) [0/1]	0
	Level_UV (Under voltage detection) [pu]	--
	Array level_UV_time () [pu sec]	--
		--
		--
		--
Over frequency Protection	ONOFF_OF (0:off/1:on) [0/1]	1
	Level_OF (Overfrequency detection) [pu]	1,03
	Array level_OF_time () [pu sec]	1,031
		0.02
		--
		--
Under	ONOFF_UF (0:off/1:on) [0/1]	1
		1

frequency Protection	Level_UF (Underfrequency detection) [pu]		0.95
	Array level_UF_time () [pu sec]	0,949	0,02
		--	--
		--	--
		--	--
Over Current Protection	ONOFF_OC (0:off/1:on) [0/1]		0
	Level_OC (Overcurrent detection) [pu]		--
	Array level_OC_time () [pu sec]	--	--
Asymmetrical Voltage Protection	ONOFF_OVNEG (0:off/1:on) [0/1]		0
	Level_OVNEG (Asymmetrical voltage detection) [pu]		--
	Array level_OVNEG_time () [pu sec]	--	--
Asymmetrical Current Protection	ONOFF_OCNEG (0:off/1:on) [0/1]		0
	Level_OCNEG (Asymmetrical current detection) [pu]		--
	Array level_OCNEG_time () [pu sec]	--	--